

Assignment 5: Pointing (20P)

Hand in in groups of two.

Goals

You can apply advanced computer vision libraries to implement pointing techniques. You can evaluate pointing techniques under varying latency conditions and visualize results.

1 Pose-Based Pointing Technique

5P

Create a new Python program called *pointing_input.py*. Use the Google MediaPipe library to detect users' hand and/or body pose. It might be necessary to counteract low camera frame rates by interpolating between frames! It should map suitable body landmarks to a pointer position and allows for clicking. Use the *pynput* library to move the mouse pointer.

Score

- (1P) Pose detection works reliably and with low latency.
- (1P) Mapping of landmarks to a pointer position works and makes sense.
- (1P) Mapping of landmarks for clicking works and makes sense.
- (1P) The application controls the mouse pointer and simulates clicks.
- (1P) Interaction feels good and works smoothly.

2 Fitts' Law Application

2P

Create a new *pyglet* program called *fitts_law.py*. The program should display a Fitts' Law study setup and log data automatically. It should be controllable with mid-air pointing, too. Use command line parameters and a config file to set important parameters, such as participant ID, the number of trials, or Fitts' Law configuration.

Score

- (1P) The Fitts' Law application works.
- (0.5P) Data is logged correctly.
- (0.5P) The application can be parametrized.

3 Steering Law Application

2P

Create a new *pyglet* program called *steering_law.py*. The program should display a Steering Law study setup and log data automatically. It should be controllable with mid-air pointing, too. Use command line parameters and a config file to set important parameters, such as participant ID, the number of trials, or Steering Law configuration.

Score

- (1P) The Steering Law application works.
- (0.5P) Data is logged correctly.
- (0.5P) The application can be parametrized.

4 Adding Latency

3P

Add an option to your applications that allows for adding latency to the pointer. Make sure, data is still logged correctly. Directly adapt the files of task 2 and 3, there is no need for additional files for this task.

Hint: There will be a session on latency next week.

Score

(3P) Latency can be added to Fitts' Law application and Steering Law application.

5 Evaluating Input Techniques

8P

Evaluate different pointing techniques for both Fitts' Law and Steering Law. Conduct the experiment with at least three participants. You and your team mate are also allowed to participate. The third one must not currently be a member of the ITT course.

Investigate at least four **input device conditions**:

- Input via pose-based pointing
- Input via mouse
- Input via mouse with additional latency of 150 ms
- Input via touchpad

For each input device condition, further investigate

- at least three different target radii for Fitts' Law
- at least three different target distances for Fitts' Law
- at least three different tunnel widths for Steering Law
- at least three different tunnel distances for Steering Law

Every condition should be repeated at least three times per participant.

Choose an appropriate procedure for your study. Describe it, capture data, and report the results of your study. Remember everything you have learned about designing good user studies in the past. Also, describe all problems that occurred during implementation of the input technique, implementing the apparatus, and conducting and analyzing results.

(2P) Methodology and study procedure are described well.

(2P) Data of at least three participants is captured appropriately.

(2P) Results are reported, compared, and discussed well.

(1P) Results are adequately visualized.

(1P) There is a section discussing problems that occurred during the assignment.